Research Areas

BIT Jaipur

The department of sciences is engaged in the following focused research areas.

Physics Research Areas

- **Plasma Science and Technology:** Atmospheric pressure glow discharge, DBD sources, Plasma medicine, Plasma nitriding, Impurity transport in tokamak plasma.
- **Material Science & Technology:** Plasma based material treatment, Materials testing and Analysis.
- **Micro-strip Antenna designs:** Microstrip Antennas and Arrays, Broadband, Ultra-Wideband, Super Ultra-Wideband Antennas and RF communication technology.

Chemistry Research Areas

- Ground Water Analysis and Environmental Pollution
- Industrial Water Pollution, Soil and Environmental Studies
- Environmental and Biological Sciences

Mathematics Research Areas

- Heat and Mass Transfer in Boundary Layer Flows
- Fluid Dynamics

Recent Research Project

Project title: Studies of High Z impurity behaviour using VUV spectral emissions from the tokamaks in IPR.

Funding Agency: BRNS, Department of Atomic Energy, Govt. of India

Sanctioned Grant: 25.6 Lakhs

Duration: 2017-2020

Conference/Workshop Organized by the Department

1. International Workshop on Cold Plasma Technology and Applications **(CPTA-2025)**, 6-8 February 2025.
2. IPR Scientific Outreach Programme on Plasma, Its Applications and Nuclear Fusion **(PLASMA GYAAN - 2022)**, 22-26 August 2022.
3. 36th National Symposium on Plasma Science and Technology **(PLASMA-2021)**, 13-15 December 2021.
4. IAEA International Workshop on Challenges in Plasma Spectroscopy for Future Research Machines **(CPS-08)**, 20-22 February 2008.
5. One day workshop on Environment Friendly Plasma Nitriding Process for Surface Hardening of Industrial Components (Commemoration of National Science Day in Rajasthan), 11 May 2008.

Webinars / Guest Lectures / Expert Talks Conducted

1. Title: Surface Plasmons: Fundamentals and Applications in Fiber Optic Sensing.
Speaker: *Dr. R K Verma, Assistant Professor, Department of Physics, Central University of Rajasthan.*
Date: 12th August 2022

2. Title: Solar Thermal Engineering for Sustainable Future.
Speaker: Dr. G D Agarwal – Professor, Mechanical Department, MNIT Jaipur, Rajasthan.
Date: 7th June 2022.